Network Discovery Detection – SOC Investigation Report
(TryHackMe Lab)

Completed by: Andrii Hurskykh
Platform: Elastic Stack (ELK)
Lab: TryHackMe — Network Discovery Detection
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Executive Summary

This report presents a Security Operations Center (SOC) investigation focused on detecting network reconnaissance activity within the *Network Discovery Detection* lab. Using Elastic SIEM and command-line log analysis, the investigation identified patterns consistent with host discovery and network scanning.

Analysis of ICMP traffic, log summaries, and raw log extraction revealed repeated probing behavior across multiple hosts. These findings strongly indicate reconnaissance activity typically associated with the early stages of a cyber attack lifecycle.

1. Introduction

Network discovery is a fundamental phase in adversarial operations. Attackers perform reconnaissance to identify active hosts, reachable services, and network topology before attempting exploitation.

From a SOC perspective, early detection of such activity provides a critical opportunity to prevent further intrusion.

This investigation demonstrates how reconnaissance can be detected using:

- SIEM-based log analysis (Elastic)
- Traffic filtering (ICMP detection)
- Command-line forensic techniques

2. Technical Environment

Component	Description
SIEM Platform	Elastic Stack (Kibana)
Data Source	Network Logs
Analysis Type	Network Threat Detection
Tools Used	Kibana, Linux CLI (awk, head)

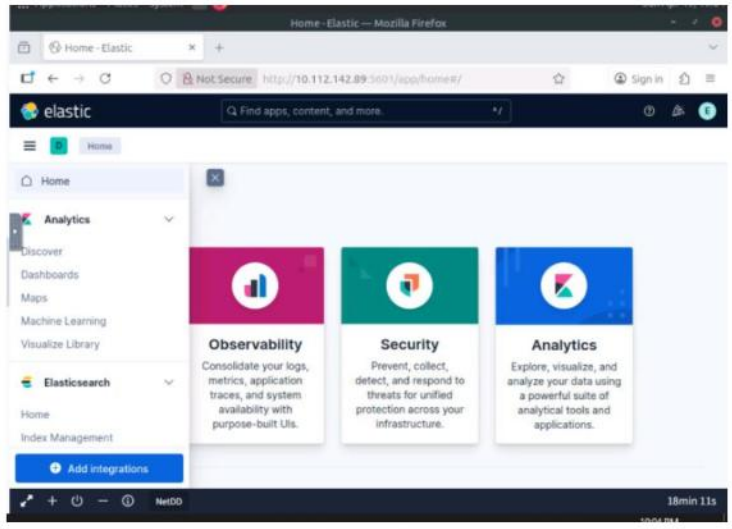
3. Initial Visibility – Elastic Dashboard

SOC Analysis:

The Elastic dashboard provides centralized visibility into ingested network logs. This confirms that log collection is functioning correctly and allows analysts to monitor network activity in real time.

SOC Insight:

The dashboard serves as the **visibility layer**, enabling analysts to pivot quickly into deeper investigations.



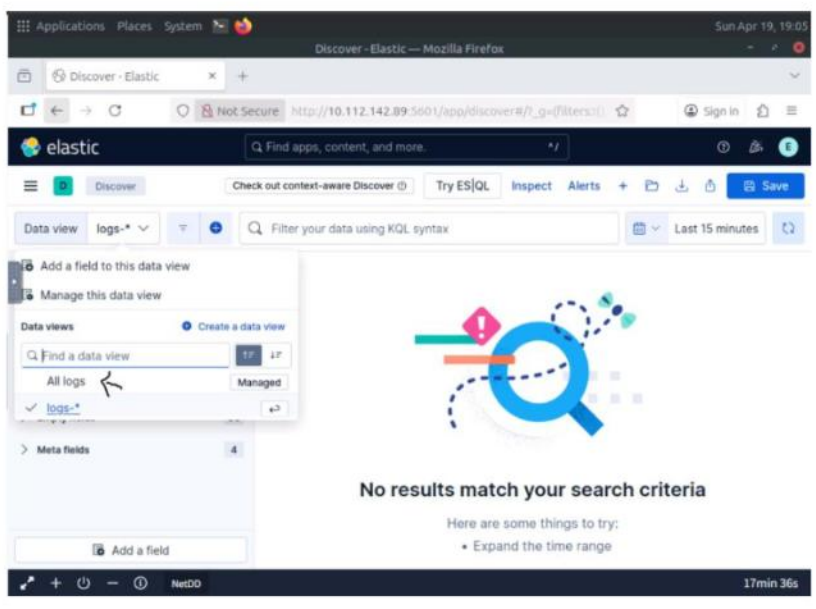
4. Baseline Log Exploration

SOC Analysis:

A broad query was executed to retrieve all available network logs. This step establishes a baseline and allows the analyst to understand the volume and structure of incoming data.

SOC Insight:

Effective investigations always begin with broad visibility before narrowing scope.



5. Log Review & Summary Analysis

SOC Analysis:

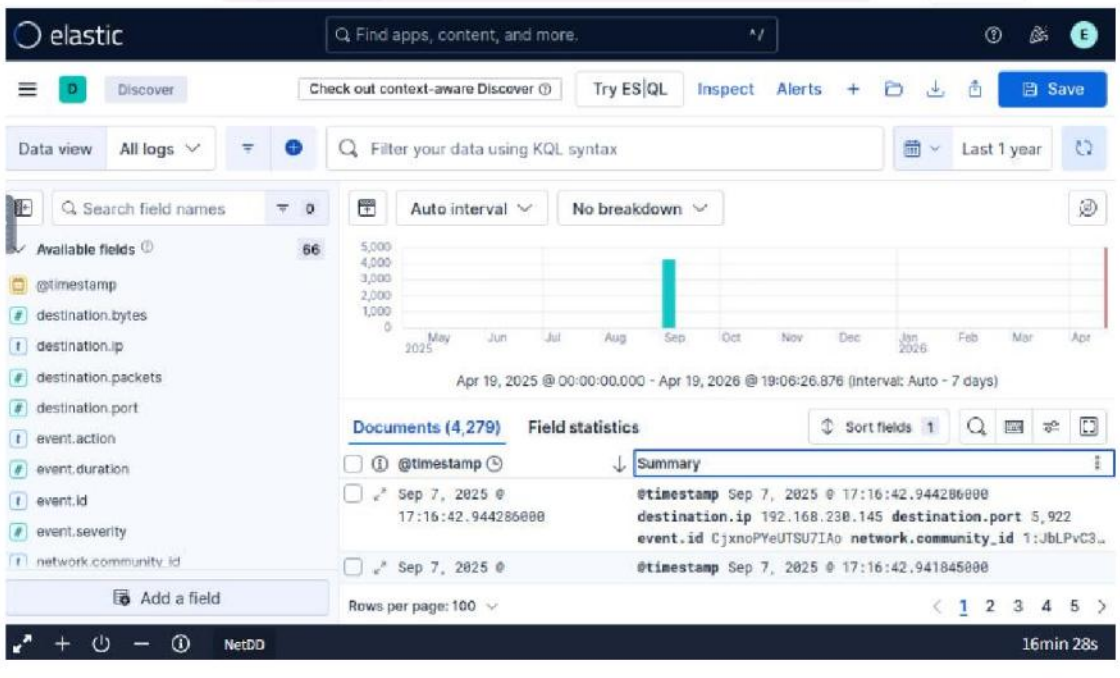
Reviewing log summaries provides insight into:

- Traffic distribution
- Event frequency
- Protocol usage

Observed patterns indicate repeated network activity that deviates from typical user behavior.

SOC Interpretation:

Abnormal repetition and structured patterns often indicate automated processes rather than legitimate human interaction.



6. ICMP Traffic Detection (Host Discovery)

Query used:

network.transport: icmp

SOC Analysis:

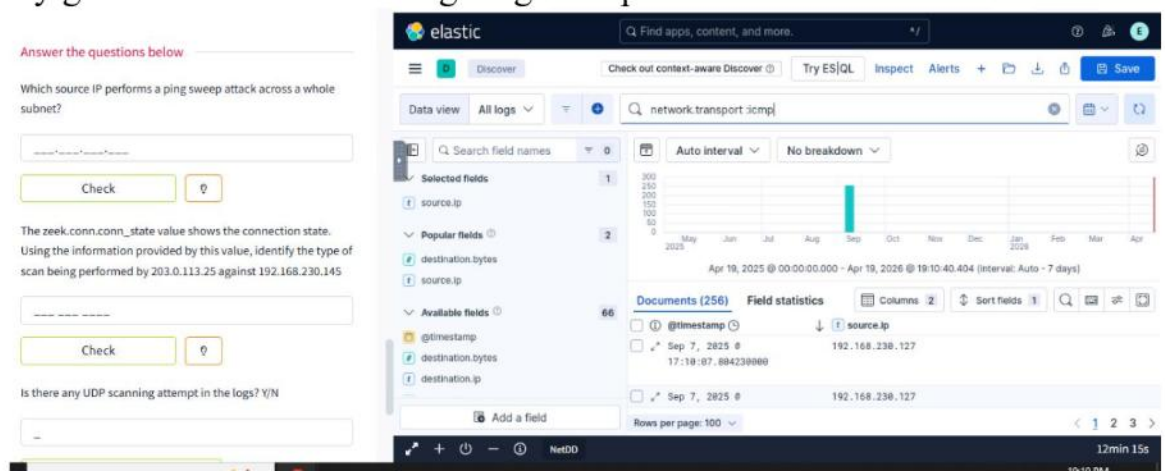
The query isolates ICMP traffic, commonly associated with ping requests. The results show multiple IP addresses being targeted.

SOC Interpretation:

This behavior is consistent with **host discovery scanning**, where an attacker sends ICMP requests to identify active systems within a network.

SOC Insight:

Legitimate users rarely generate ICMP traffic targeting multiple hosts in rapid succession.



7. Command-Line Log Forensics

Command used:

```
head -n 2 <log_file>
awk -f ‘,’  ‘{print $3” ,$4” ,$5” ,$6”} log-session-2.csv >test.csv
```

SOC Analysis:

Command-line tools were used to extract and analyze raw log data. This approach allows deeper inspection beyond SIEM dashboards.

Key Observations:

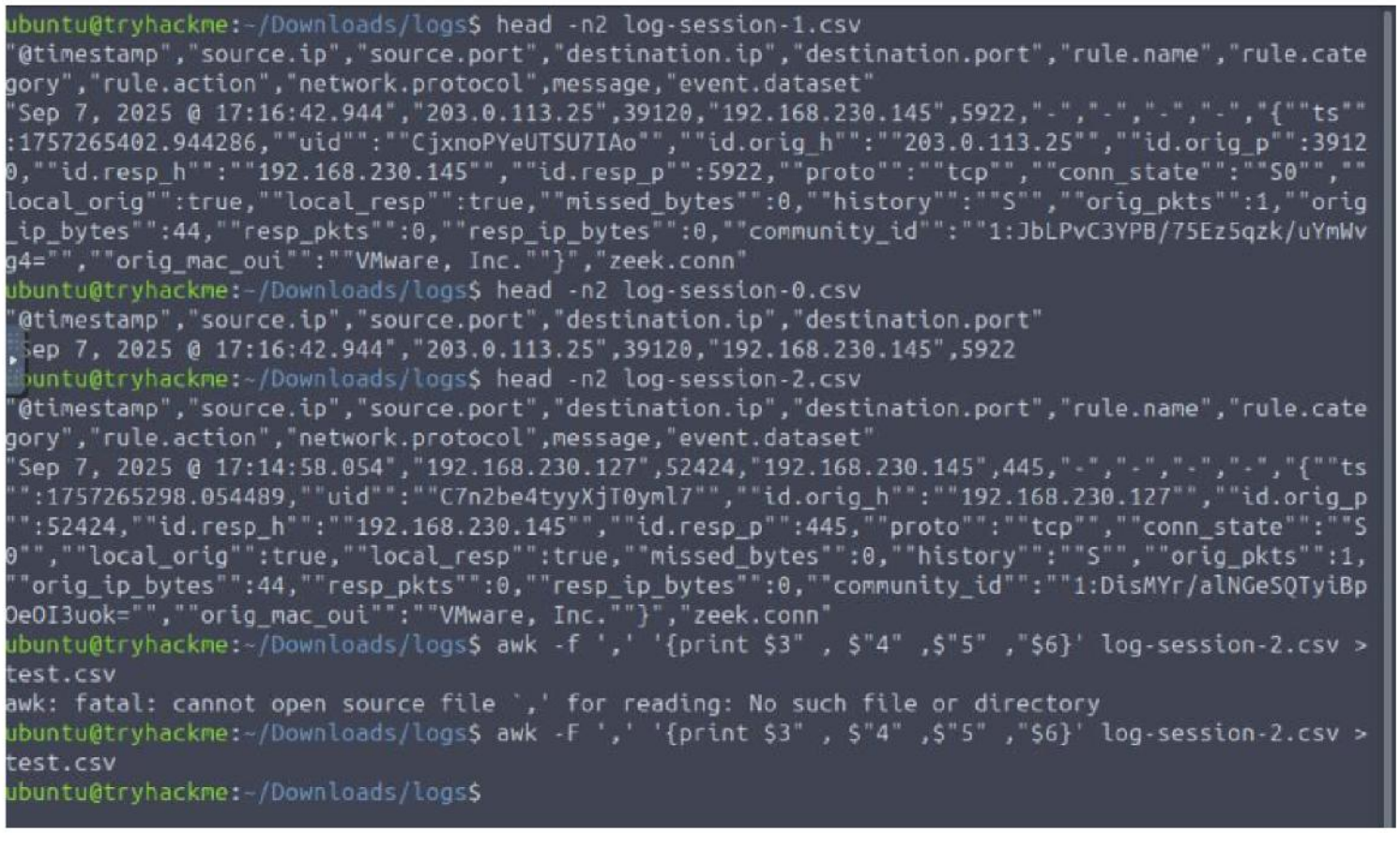
- Structured log entries
- Repeated source/destination patterns
- Consistent traffic formatting

SOC Interpretation:

Manual log parsing confirms the presence of **systematic scanning behavior**, reinforcing findings from SIEM analysis.

SOC Insight:

Skilled analysts validate SIEM findings using raw log analysis for higher confidence.



8. Threat Analysis & Behavioral Assessment

The investigation identified:

- Repeated ICMP requests
- Multiple target IP addresses
- Structured and automated traffic patterns

SOC Interpretation:

These indicators strongly suggest:

- Automated scanning tools
- Network reconnaissance activity
- Pre-attack behavior

This aligns with the reconnaissance phase of an intrusion.

9. MITRE ATT&CK Mapping

Behavior	Technique	ID
Host Discovery	Active Scanning	T1595
Network Scanning	Network Service Discovery	T1046

10. Detection Methodology

The investigation followed a structured SOC workflow:

- Establish visibility via SIEM
- Retrieve all logs (baseline)
- Analyze summaries and patterns
- Filter specific protocol (ICMP)
- Validate findings via CLI analysis
- Classify behavior

SOC Insight:

This layered approach increases detection accuracy and reduces false positives.

11. Response Recommendations

Immediate Actions

- Monitor source IP generating ICMP traffic
- Block suspicious scanning sources

Short-Term Improvements

- ❖ Configure alerts for ICMP scanning thresholds
- ❖ Implement IDS rules for reconnaissance detection

Long-Term Strategy

- ❖ Integrate SIEM with network detection tools
- ❖ Deploy behavioral analytics solutions
- ❖ Develop threat-hunting procedures

12. Final SOC Assessment

The investigation identified clear indicators of network reconnaissance activity through ICMP-based host discovery.

The combination of:

- ❖ SIEM log analysis
- ❖ Protocol filtering
- ❖ Command-line validation

provides strong evidence of scanning behavior.

Final Verdict:

The activity is classified as **malicious reconnaissance**, representing the initial stage of a potential attack. Immediate monitoring and mitigation actions are recommended.

13. Conclusion

This investigation highlights the importance of detecting early-stage attacker behavior. By identifying reconnaissance activity, SOC teams can proactively defend networks before exploitation occurs.

The ability to combine SIEM analysis with manual log inspection is a critical skill for modern SOC analysts.